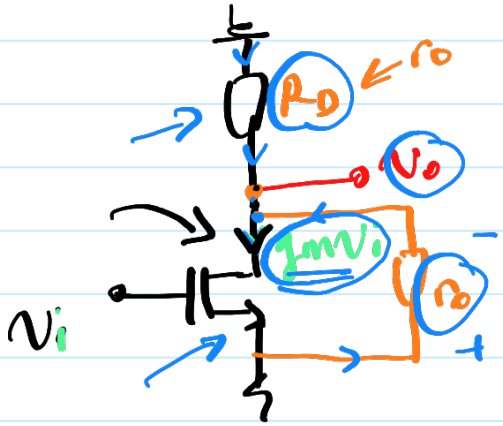
Reading

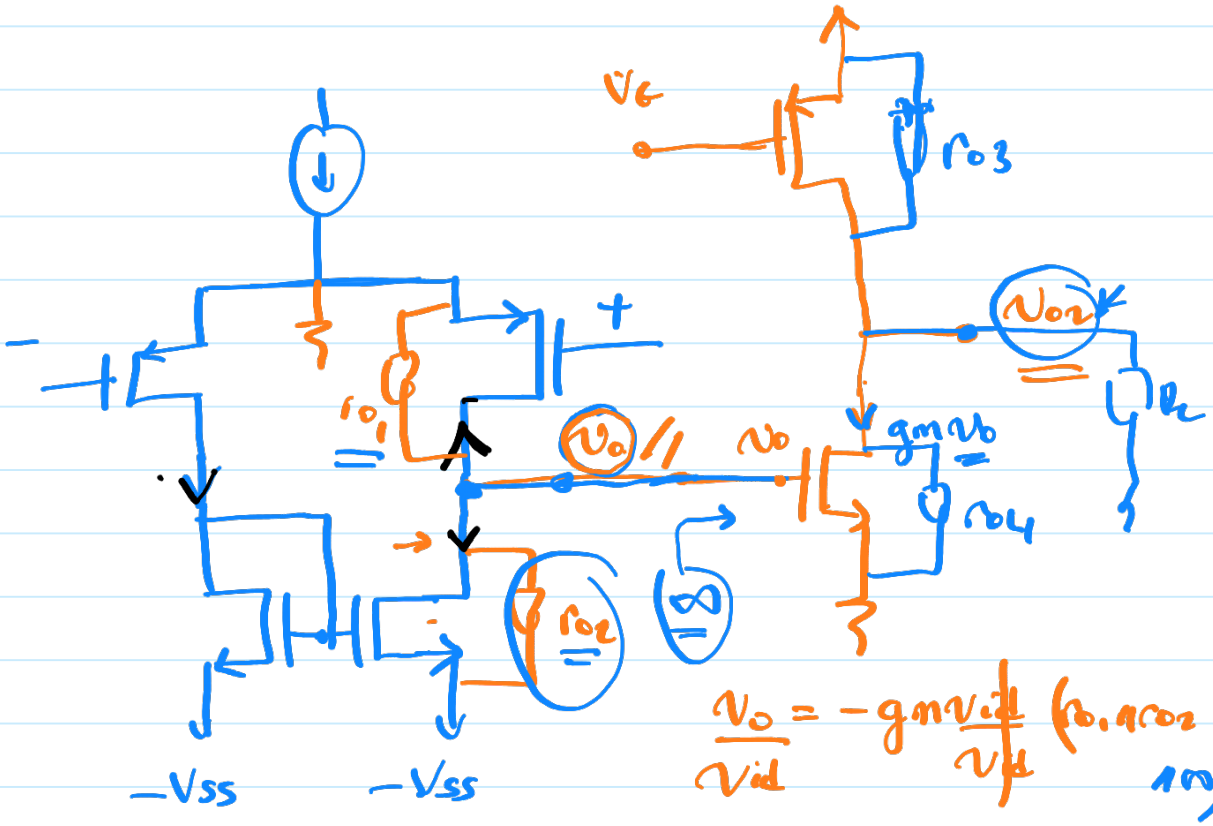
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 9.6 Multistage Amplifiers
- Chapter 10.1 High-Frequency Transistor Model

A Two Stage CMOS OPAMP



$$v_o = -g_m v_i (R_D \| r_o)$$



$$\frac{v_o}{v_{id}} = -g_m \frac{v_{id}}{2} (r_{o1} \| r_{o2} \| \infty)$$

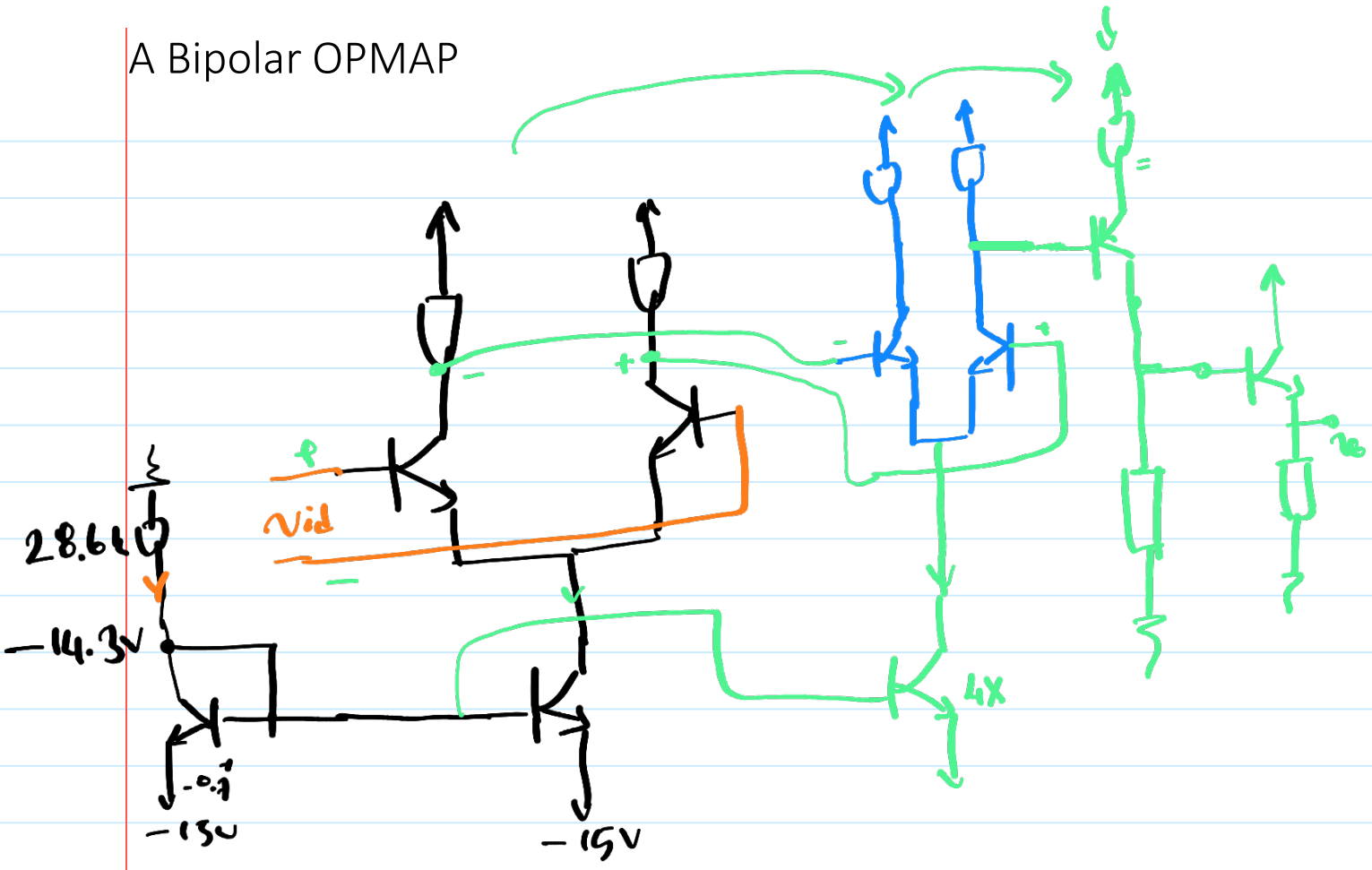
$$A_1 = -g_m (r_{o1} \| r_{o2} \| \infty)$$

$$A_2 = \frac{v_{o2}}{v_o} = ?$$

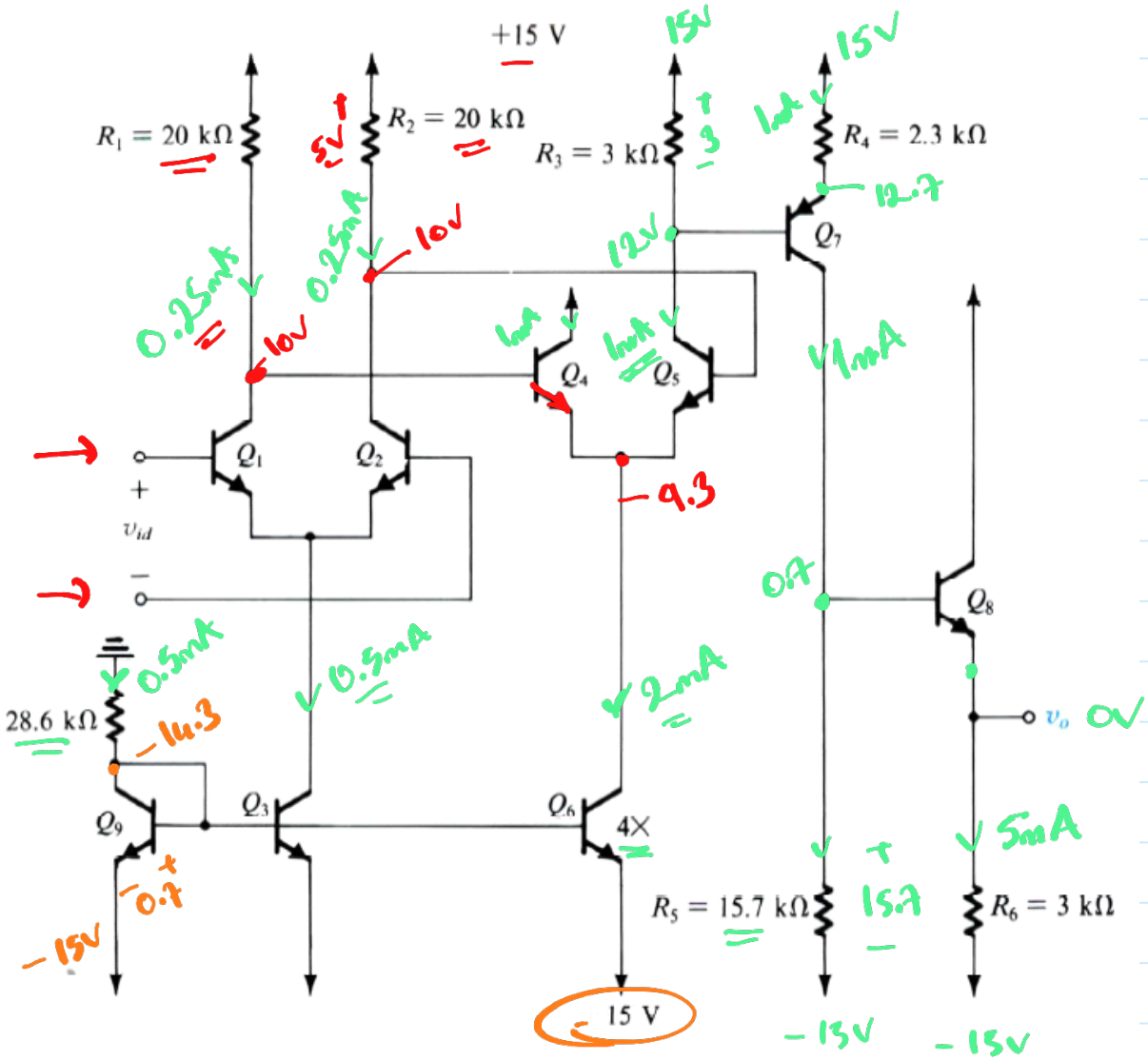
$$\frac{v_{o2}}{v_o} = g_{m4} v_o (r_{o3} \| r_{o4} \| R_L) = A_2$$

$$A = \frac{v_{o2}}{v_{id}} = \underbrace{\frac{v_{o2}}{v_o}}_{A_2} \cdot \underbrace{\frac{v_o}{v_{id}}}_{A_1} = A_2 A_1$$

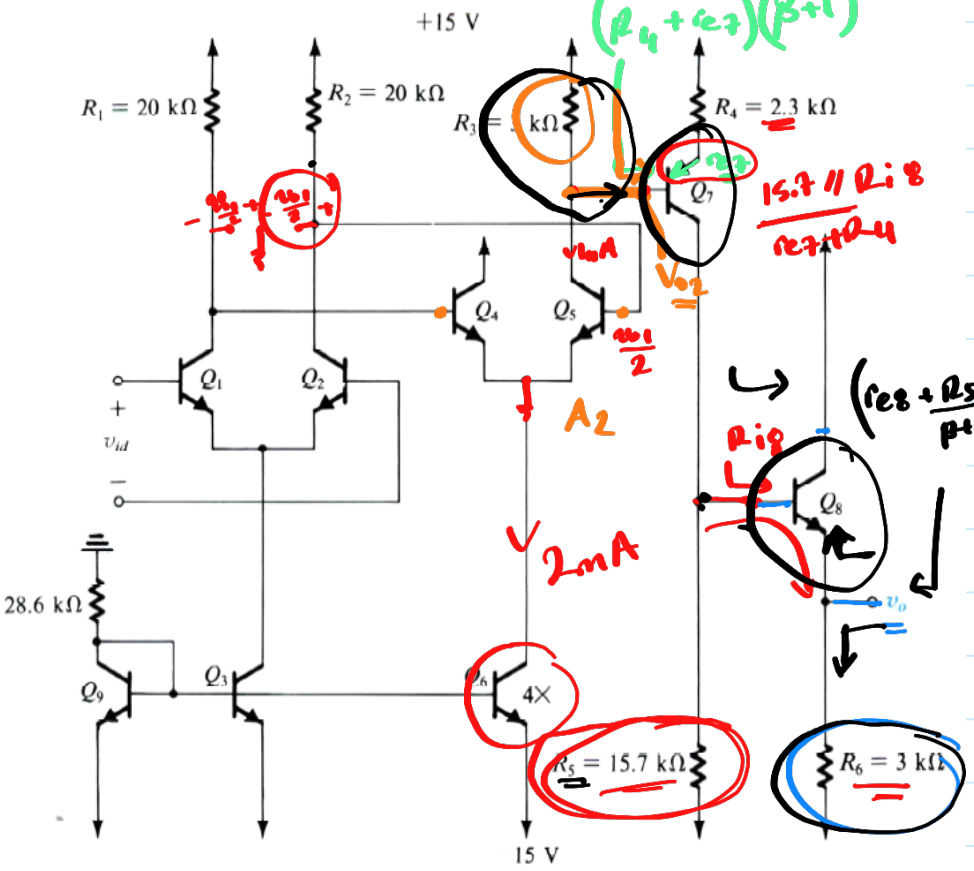
A Bipolar OPMAP



Example


+15 V
$R_1 = 20$ kΩ
$R_2 = 20$ kΩ
$R_3 = 3$ kΩ
$R_4 = 2.3$ kΩ
Q_1
Q_2
Q_3
Q_4
Q_5
Q_6
Q_7
Q_8
Q_9
v_{id}
+
−
28.6 kΩ
$R_5 = 15.7$ kΩ
$R_6 = 3$ kΩ
4×
15 V
v_o
0V
−15V
−15V
−15V

Example



$r_{e1} = r_{e2} = 100\ \Omega$

$r_{\pi 1} = r_{\pi 2} = 10\ k\Omega$

$R_{in} \cong 2\ M\Omega$

$r_{o4} = r_{o5} = 25$

$r_{\pi 4} = r_{\pi 5} = 2.5\ k$

$g_m = \frac{\alpha}{r_e} = \frac{1}{100}$

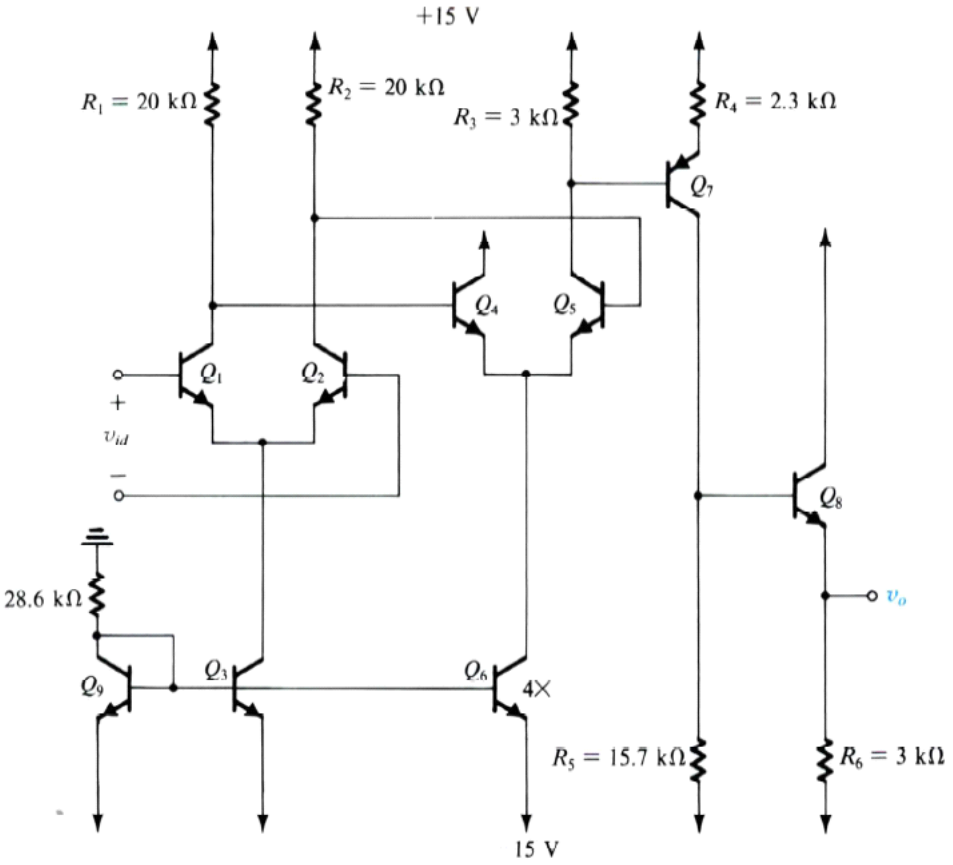
$100\ mA/V$

$A_2 = \frac{R_6}{r_{e8} + R_6}$

$A_1 = \frac{20k \parallel 2.5k}{r_e}$

$A_1 = 22\ V/V$

Example


+15 V
$R_1 = 20$ kΩ
$R_2 = 20$ kΩ
$R_3 = 3$ kΩ
$R_4 = 2.3$ kΩ
Q_7
Q_4
Q_5
Q_1
Q_2
+
v_{id}
−
Q_8
28.6 kΩ
v_o
Q_9
Q_3
Q_6
4×
$R_5 = 15.7$ kΩ
$R_6 = 3$ kΩ
15 V

Frequency Response

The High Frequency Model of MOSFET

The MOSFET Unity Gain Frequency

The High Frequency Model of BJT

The BJT Unity Gain Frequency

Template

Template

Template